Question 1 (50 marks) (CLO2-PLO2-C4)

The RLC circuit has a 10 Ω pure resistor, a coil of inductance 2 H with negligible resistance, and a 250 μF capacitor connected in series, as shown in Figure 1. The circuit is supplied with an instantaneous a.c. voltage, $v_1(t)$, of the following:

$$v_1(t) = 10\ cos\ (50t\ +\ 30°)$$

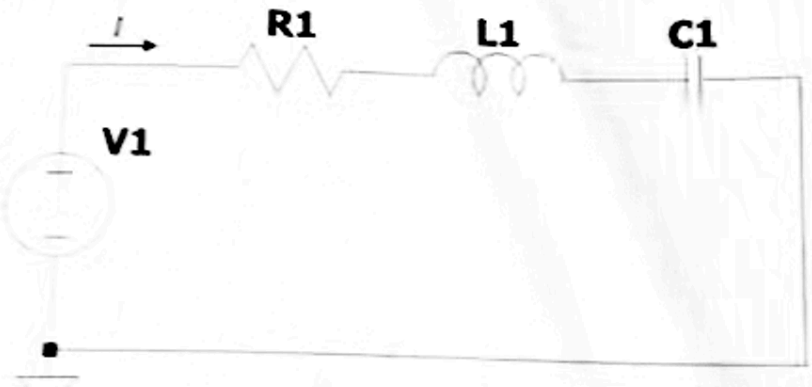


Figure 1

a) Calculate the total impedance, Z_T, of the series connected RLC components. Express it in complex form.

[12 marks]

b) Calculate the current, I. Express it in complex form. **[12 marks]**

c) Determine the magnitude and phase angle of the current, I. **[8**

d) Determine the phase angle of the circuit. **[6 marks]**

e) Illustrate the phasor diagram for I, V, and Z_T. Indicate the magnitude and phase angle of all the phasors. **[12 marks]**

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Question 2 (50 marks) (CLO2-PLO2-C4)

The RLC circuit has a 10 Ω pure resistor, a coil of inductance of 2 H with negligible resistance, and a 1 mF capacitor connected in parallel, as shown in Figure 2. The circuit is supplied with an instantaneous a.c. voltage, $v_1(t)$, of the following:

$$v_1(t) = 10\cos(50t + 30°)$$

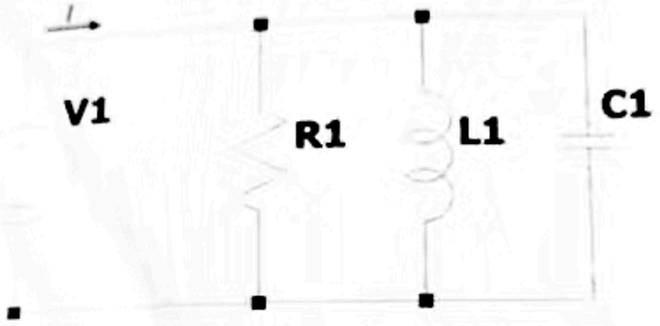


Figure 2

a) Determine the total impedance, Z_T, of the parallelly connected RLC components. Express it in complex form. **[12 marks]**

b) Calculate the current I. Express the current in complex form. **[12 marks]**

c) Determine the magnitude and phase angle of the current, I. **[8 marks]**

d) Determine the phase angle of the circuit.

[6 marks]

e) Illustrate the phasor diagram for V, I, and Z_T. Indicate the magnitude and phase angle of all the phasors.

[12 marks]